

ARTESIA, NM, USA

WMO: 722676

Lat: 32.853N

Lon: 104.468W

Elev: 1079

StdP: 89.01

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 03035

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-7.3	-4.9	-18.7	0.8	4.8	-17.1	0.9	7.5	15.0	14.3	13.0	13.3	2.7	160	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	38.7	18.4	37.5	18.8	36.3	18.9	22.8	30.1	22.2	29.8	21.7	29.4	5.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.3	18.3	24.1	20.9	17.8	23.8	19.8	16.6	23.6	73.3	30.1	70.8	29.4	68.4	29.0	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.2	9.4	DB	-10.9	41.3	3.7	0.9	-13.6	41.9	-15.8	42.5	-17.9	43.0	-20.6	43.7	(4)
(5)			WB	-12.0	23.5	3.4	1.1	-14.4	24.3	-16.4	24.9	-18.3	25.5	-20.7	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	6.4	8.8	13.1	17.6	22.1	27.1	27.8	27.4	23.5	17.4	11.1	6.1		
	DBStd	8.84	4.88	5.08	4.57	4.26	4.01	2.49	2.34	2.31	3.09	4.36	4.68	4.74		
	HDD10.0	416	132	73	23	4	0	0	0	0	0	5	42	137		
	HDD18.3	1551	370	267	168	61	14	0	0	0	3	68	221	379		
	CDD10.0	3123	20	40	119	233	376	512	553	539	406	234	74	16		
	CDD18.3	1216	0	1	6	41	132	262	294	281	159	39	3	0		
	CDH23.3	15898	8	46	239	781	1899	3521	3722	3366	1708	528	77	3		
	CDH26.7	8447	1	6	58	298	984	2089	2122	1859	824	192	13	0		
(14) Wind	WSAvg	4.1	3.8	4.4	4.7	5.2	4.9	4.6	3.8	3.4	3.7	3.8	3.8	3.8		
(15) Precipitation	PrecAvg	318	11	11	14	16	34	35	49	39	44	27	15	15		
	PrecMax	550	38	45	39	122	107	82	113	109	119	76	94	78		
	PrecMin	201	0	0	2	1	3	0	7	0	4	1	0	0		
	PrecStd	102	10	12	11	26	33	24	32	28	31	23	25	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	27.5	30.2	33.8	38.0	40.9	40.0	39.0	37.4	33.0	28.0	23.0		
		MCWB	10.3	11.9	12.9	14.4	17.1	17.8	19.7	19.4	19.6	15.8	12.8	9.9		
	2%	DB	22.1	24.7	28.0	32.2	36.2	38.9	37.9	37.5	35.0	31.2	26.0	21.0		
		MCWB	9.7	10.6	11.9	13.6	16.1	17.5	19.7	19.8	18.7	16.4	12.5	9.5		
	5%	DB	18.8	22.3	26.2	30.0	34.0	37.6	37.2	36.3	33.0	28.8	22.8	17.7		
		MCWB	8.5	9.7	11.1	13.0	15.0	17.7	19.7	19.9	18.1	15.0	11.4	8.0		
	10%	DB	16.1	19.0	23.7	27.9	32.4	36.1	35.2	34.8	32.1	27.0	20.1	15.0		
		MCWB	6.8	8.4	10.3	12.5	15.0	17.9	19.9	19.9	18.3	15.0	10.5	6.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	13.3	16.5	18.3	21.1	23.5	23.6	23.4	22.7	21.3	17.1	12.4		
		MCDB	20.8	22.8	21.8	24.4	29.5	30.8	31.8	29.7	28.0	24.9	20.6	18.0		
	2%	WB	10.5	11.9	14.2	16.6	19.8	22.3	22.8	22.7	21.7	19.9	14.8	10.4		
		MCDB	18.8	21.8	22.4	22.9	28.2	29.9	30.8	29.6	28.0	23.7	20.8	18.4		
	5%	WB	9.1	10.5	12.8	15.2	18.7	21.3	22.1	22.0	20.9	18.4	12.8	8.9		
		MCDB	17.3	20.4	21.9	24.3	27.0	29.5	30.4	29.5	27.3	23.7	19.8	16.9		
	10%	WB	7.5	9.1	11.4	14.0	17.6	20.6	21.6	21.5	20.2	17.0	11.4	7.4		
		MCDB	14.8	18.1	21.1	24.9	25.5	29.2	29.9	29.2	26.8	22.3	19.2	13.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	16.3	16.8	17.1	16.4	15.9	13.5	13.6	13.7	14.8	15.3	14.4		
		MCDBR	19.8	20.9	20.9	19.9	19.2	18.6	16.1	15.8	16.6	18.7	19.1	18.9		
	5% WB	MCWBR	10.6	10.6	9.8	8.9	8.1	6.5	4.8	4.7	5.7	7.8	9.4	10.1		
		MCWBR	17.8	19.2	17.6	16.6	16.0	14.7	13.3	12.9	13.2	13.2	15.8	17.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.251	0.260	0.278	0.308	0.327	0.353	0.383	0.363	0.337	0.291	0.269	0.257		
		taud	2.614	2.592	2.536	2.429	2.434	2.404	2.368	2.416	2.445	2.581	2.572	2.590		
	Ebn at Noon	Ebn at Noon	992	1011	1008	980	957	926	897	913	928	959	958	965		
		Edh at Noon	75	86	98	115	115	119	123	115	107	86	78	73		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	3.51	4.45	5.72	7.03	7.59	7.75	7.26	6.63	5.74	4.67	3.68	3.09		
	RadStd	RadStd	0.30	0.38	0.36	0.42	0.42	0.34	0.34	0.27	0.37	0.43	0.32	0.24		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	-0.72	N/A	N/A	N/A	N/A	N/A		
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page